

Lab 1: Requirements Table

Problem Statement #23: Hyperlocal Courier Dispatch & Tracking Engine

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall match a courier request with the nearest active delivery rider within a 3 km radius.	High	Pass: Nearest active rider within 3 km receives the request. Fail: An offline or farther rider is assigned when a nearer active rider is available.	This helps reduce pickup time and makes local delivery faster.
FR-002	Functional	The system shall allow the Sender Client to create a courier request by entering the pickup location, delivery location and parcel details.	High	Pass: Request is created only after all required details are entered and a request ID is generated. Fail: Request is created with required details missing.	The sender needs to provide the delivery details before a rider can be assigned.
FR-003	Functional	The system shall allow the Delivery Rider to accept or reject a courier request assigned to them.	High	Pass: Accepting changes the status to Accepted and notifies the sender; rejecting starts reassignment. Fail: A rejected request remains assigned to the same rider.	A rider may not always be available to take a delivery.
FR-004	Functional	The system shall generate a delivery route for the rider and optimize the order of stops when multiple deliveries are assigned.	High	Pass: The route includes all assigned stops in an ordered route. Fail: Any assigned pickup or delivery stop is missing.	A better route can save travel time and reduce unnecessary distance.
FR-005	Functional	The system shall verify an OTP at the destination before the courier can be marked as delivered.	High	Pass: A correct OTP changes the status to Delivered. Fail: An incorrect OTP allows the delivery to be completed.	OTP verification confirms that the parcel reached the correct recipient.
NFR-001	Non-Functional (Performance & Security)	Delivery status and live GPS updates from riders shall reach the Sender Client in under 2 seconds.	High	Pass: GPS and status updates reach the Sender Client in under 2 seconds during peak-load testing. Fail: Updates take 2 seconds or more or are insecure.	The sender should be able to track the delivery without noticeable delay.
NFR-002	Non-Functional (Availability / Reliability)	The system shall maintain at least 99.5% availability during normal operation.	Medium	Pass: Monthly system availability is at least 99.5%. Fail: Monthly availability falls below 99.5%.	The system should stay available for booking, dispatch and tracking.